

## Direct and indirect proofs



- 1 Sari wants to prove that  $(a + b)^2 \geq 2ab$  for all integers  $a$  and  $b$ . Her proof is shown below:

$$\begin{aligned}(a + b)^2 &= a^2 + b^2 + 2ab \\ &\geq 2ab \quad \text{since } \mathbb{I}\end{aligned}$$

What should she write in the  $\mathbb{I}$  to complete her proof?

- 2 Prove that  $(x + y)^2 - 9(x - y)^2 = 4(2x - y)(2y - x)$ .
- 3 Use the method of completing the square to prove that  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  for all quadratics of the form  $ax^2 + bx + c = 0$ .
- 4 If  $N$  is an even integer, prove that  $\frac{N^2}{2}$  is an even integer.
- 5 Prove that angle sum of a straight line is  $180^\circ$ .
- 6 Prove that the sum of two consecutive numbers is odd.
- 7 Prove that  $(a + 2)^2 - (a - 2)^2$  is divisible by 8 for any positive whole number  $a$ .
- 8 Prove that the sum of two consecutive odd numbers is an even number.
- 9 Prove that the distance between two points  $(a, b)$  and  $(c, d)$  is given by  $\sqrt{(c - a)^2 + (d - b)^2}$ .
- 10 Prove that if we subtract 1 from a positive odd square number, the answer is always divisible by 8.
- 11 Prove that  $0.\dot{1}\dot{8}$  is a rational number.